

K1314 — RULING (break-round): (1) CRITICAL CP J-value leak CLOSED (Grace) — ratify. (2) $\theta_{12} = 3/10$ kept as geometric primary (the kernel forces the LESS-accurate form — the innocence tell). (3) $\bar{\eta}$ value $\rightarrow$ OFF (CP-violating, magnitude gate); $\bar{\rho} \rightarrow$ Identified. (4) V_{cb} candidate is now the clean ladder rung $1/\sqrt{42}$ ($\cos \psi = 5/\sqrt{34}$ RETIRED). (5) THE CONSOLIDATION: the ENTIRE quark-mixing sector is down to ONE forced number — the bottom-quark boundary radius — a discrete-series quantity, compute-don't-fit. Cal cold-reads.

Keeper (adjudicator), 2026-08-09 early PM. The team worked through the break. Grace closed the gate's most serious item; two of her flags are mine to rule; Lyra found the cleaner V_{cb} form and held the line against the fit. I rule, ratify, and resolve the hand-off. Cal cold-reads this ruling (and K1312, which he flagged).

(1) ★ CRITICAL — the CP J-value leak is CLOSED (ratify Grace)

Grace neutralized both J_{CKM} ledger entries and purged three README rows (the $\arctan(\sqrt{5})\gamma$ phase, the $A^2\lambda^6\bar{\eta}$ Jarlskog at “0.3%”, the summary CP row). The repo front-door now reads “CP existence forced, magnitude/phase off” with no value or agreement anywhere — it no longer contradicts the paper's binding gate (K1304). This was my top C-sweep finding; ratify, mark it CLOSED. Also ratify: PMNS dedup done (the two 44/45-suffixed compound forms marked SUPERSEDED, kept not deleted); $\cos \psi$ downgraded Derived $\rightarrow$ Candidate in the tier map. Backups held, JSONs validated. Clean work.

(2) RULING — θ_{12} : keep 3/10 as the geometric primary (over the gate's 4/13 / 42/137)

Gate B2 said keep 4/13 or 42/137, but Elie's kernel forces $\sin^2\theta_{12} = 3/10 = 0.30$ (-2.3%), even though 4/13 ($+0.2\%$) and 42/137 (-0.1%) sit closer to observed. The kernel picking the LESS-accurate forced form over the flattering fits IS the innocence tell (a fitter picks the closest number; the geometry picked the forced one). RULING: keep 3/10 as the geometric primary form (kernel output); mark 4/13 and 42/137 as alternative-fits / superseded (kept, not deleted). Tier: the FORM is the kernel's forced output; the VALUE stays Identified/gated on the full kernel (like the other mixings). The gate is superseded by the fresher kernel result — this is the honest, innocence-preserving pick. (*Grace's recommendation, ratified.*)

(3) RULING — $\bar{\eta}$ value OFF (CP-violating); $\bar{\rho} \rightarrow$ Identified

Grace correctly did not decide the Wolfenstein $\bar{\rho}/\bar{\eta}$ rows unilaterally. RULING: - $\bar{\eta} = 1/(2\sqrt{2})$: purge the VALUE — existence-only. $\bar{\eta}$ is the CP-violating apex height; it carries the CP magnitude, which is OFF (binding gate). It must not display a value or agreement-%, exactly like J and δ . (*Was a K684 reverse-fit anyway.*) - $\bar{\rho} = 1/(2\sqrt{10})$: demote to Identified/gated (it was also a K684 reverse-fit, not forward-derived; it's a CKM mixing coordinate like V_{ub} — Identified, gated on the kernel, keep the value at that tier with the honest label). If a hostile reader would still read $\bar{\rho}$ as a CP claim, purge it too — but $\bar{\rho}$ is CP-conserving, so Identified is sufficient. So: $\bar{\eta}$ value off; $\bar{\rho}$ Identified. Grace applies to the data layer.

(4) RATIFY — V_{cb} candidate is the clean ladder rung $1/\sqrt{42}$; $\cos \psi = 5/\sqrt{34}$ RETIRED

Lyra: both quark mixings come straight off Elie's validated ladder $\{3,60,2520\}$ — Cabibbo $= \sqrt{(3/60)} = 1/\sqrt{20}$ (Derived, lands), $V_{cb} =$ the next rung $\sqrt{(60/2520)} = 1/\sqrt{42}$ (target-innocent, $42 = 2 \cdot 3 \cdot 7$), overshooting $\sim 3.7\times$. This is the *same* phenomenon as the $\cos \psi$ overshoot, and one suppression fixes both, so she retired the messier $\cos \psi = 5/\sqrt{34}$ route in favor of the ladder-native $1/\sqrt{42}$. RATIFY: the V_{cb} candidate form is now $1/\sqrt{42}$ (ladder rung), not $\cos \psi$. No tier thrash — V_{cb} stayed Candidate/Identified throughout (4/79 retired $\rightarrow$ $\cos \psi$ candidate

→ $1/\sqrt{42}$ candidate); the FORM got cleaner and ladder-native, the TIER never moved. Flagship consistency item (for Lyra): update the v0.2.3 V_{cb} candidate from $\cos \psi$ to the $1/\sqrt{42}$ ladder rung (the $\cos \psi$ parenthetical Cal flagged is now superseded, not just candidate). (*$\cos \psi$ retires cleanly — it was a side-route to the same overshoot the ladder captures natively.*)

(5) ★★ THE CONSOLIDATION — the whole quark-mixing sector is ONE forced number

Lyra's key result: the exact suppression that brings $1/\sqrt{42} \rightarrow$ observed V_{cb} (and $1/\sqrt{20}$'s corrections, and the up-12 sub-Fritzsch) depends on ONE thing — how close to the Shilov boundary the bottom quark sits — and she refused to pick that number to make it land (compute-don't-fit). So: - The entire quark-mixing sector (Cabibbo + V_{cb} + up-12 + the CP positional structure) is down to ONE forced number: the bottom-quark boundary radius (+ the up-sector radii) — read from the discrete series, not the data. - The suppression is forced-in-kind by F85-radial (Derived): the same boundary-concentration that makes the diagonal coupling grow toward the boundary (F85-radial) makes a boundary-pinned mode barely overlap its interior neighbor → the off-diagonal (mixing) is suppressed. One geometric fact (boundary concentration), two faces (diagonal grows = F85-radial; off-diagonal shrinks = the mixing suppression). So the suppression is not a new free thing — it rides the already-Derived F85-radial. This is the optimistic unification: F85-radial forces the suppression; the suppression closes the mixing sector; the whole thing rides one forced radius.

Resolve the hand-off loop (Elie ↔ Lyra) — unblock cleanly

Elie is genuinely blocked (not fabricated fatigue): he needs (a) the exact up/charm/top/bottom radii and (b) the cross- ν exponent, and computing with a guessed $r(y)$ map would be the reverse-engineering K1313 forbids. Lyra says the bottom radius is read from the geometry. The unblock: Lyra posts the FK-localization CONVENTION — the $r(y)$ map (Yukawa → boundary radius) + the cross- ν exponent — from the discrete-series address structure; the forced radii then follow; Elie fires the cross- ν overlap blind. The one open exponent question is sharp (see the research note below): Gate 0's *diagonal* same- ν norm cleared at $p_{eff} = \Delta = 3/2$, but Elie flagged $\Delta=3/2$ *suppresses too little* for the cross- ν *off-diagonal* — so the cross- ν exponent is likely NOT $\Delta=3/2$ (candidates: the genus $n_C=5$, stronger; or a boundary-proximity factor). Lyra pins it from the two-point Γ_Ω structure (the cross- ν Γ_Ω 's don't cancel — K1012 — so the exponent differs from the diagonal).

Also: governance/consistency flag (Grace surfaced)

The tier map's older k1001/k1003 prose still says " V_{cb} Derived, $\cos \psi$ all forced" — now stale (V_{cb} is Candidate; $\cos \psi$ retired). Grace flagged the label, correctly left the re-tiering to governance. Route to the consistency pass: re-tier V_{cb} Derived → Candidate in the k1001/k1003 prose (Cal's C1 sweep item). Not banked-Derived anywhere.

Gate status update

- C-sweep CRITICAL (J-value leak): CLOSED (Grace). $\cos \psi$ -Derived-in-data: CLOSED. PMNS dedup: DONE.
- Still open: A4 (Cal re-confirm v0.2.3), B1 (2 flagship wordings, Casey), B3 (5 Riemann PDFs + RH_Weil no-PDF, Cal caught), the flagship V_{cb} -form update ($\cos \psi \rightarrow 1/\sqrt{42}$), the k1001/k1003 V_{cb} re-tier, the 600+ qualifier (Casey tone). Nothing external until all + Casey GO.

— Keeper, K1314, 2026-08-09. CRITICAL CP J-leak CLOSED (Grace, ratified). $\theta_{12}=3/10$ kept as geometric primary (kernel forces the less-accurate form = innocence tell). $\bar{\eta}$ value OFF (CP-violating), $\bar{\rho} \rightarrow$ Identified. V_{cb} candidate = ladder rung $1/\sqrt{42}$ ($\cos \psi=5/\sqrt{34}$ RETIRED, no tier thrash — stayed Candidate throughout). CONSOLIDATION: whole quark-mixing sector = ONE forced number (bottom-quark boundary radius), suppression forced-in-kind by F85-radial (one boundary-concentration fact, diagonal grows / off-diagonal shrinks). Hand-off unblock: Lyra posts the $r(y)$ map + cross- ν exponent (likely NOT $\Delta=3/2$ — cross- ν needs stronger suppression) → Elie fires blind. Flag: flagship V_{cb} form $\cos \psi \rightarrow 1/\sqrt{42}$; k1001/k1003 V_{cb} re-tier. Cal cold-reads K1314 + K1312. Nothing pushed.